

**Investigating the Effects of Diversification Methods in Generating Training
Partners for Human-AI Cooperation**

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Table of Contents

Abstract.....	4
1. Introduction	5
1.1 The Human-Artificial Intelligence Cooperation Problem	5
1.2 Autonomous Vehicles as a Prime Example	5
1.3 Human-AI Cooperation as a Challenge	5
1.4 Solutions for Human-AI Cooperation	6
1.5 Limitations of Current Cooperative Agents.....	11
1.6 Proposed Research.....	12
2. Methods and Materials	12
2.1 Generating the Training Sets.....	13
2.2 Normalizing the Training Sets for Competency	13
2.3 Normalized Training Sets Measured in Diversity	14
2.4 Training Generalists with Training Sets.....	14
2.5 Generating the Held-Out Set	14
2.6 Normalizing the Held-Out Sets for Incompetency.....	14
2.7 Testing Generalists for Performance with the Held-Out Set.....	15
2.8 Testing Training Sets for Performance with the Held-Out Set	15
2.9 Data Validation via Joint Diversity	15
3. Data, Results, and Analysis	15
3.1 Diversity of Training Sets	15
3.2 Performance of Training Sets.....	18
3.3 Generalist Performance	19
3.4 Data Validation	21
4. Discussion and Future Research	22
4.1 Summary	22
4.2 Applications	23
4.3 Limitations	23
4.4 Future Work.....	24

Abstract

AI agents often fail at cooperating with humans, specifically with the diverse, suboptimal strategies that humans employ, due to traditional training methods such as self-play leading to strict conventions that only work under the assumption that all parties follow them. This issue of a lack of flexibility, formally robustness, has been researched, and diversification methods in the creation of training agents has proven some success. By incentivizing training agents to act in unpredictable and slightly suboptimal ways, human behavior and diversity can be emulated during training. However, these diversification methods have often not been fine-tuned, and the effects of these methods are not well-understood. Various populations of training agents with various XPMin values were created. The diversity and viability of these populations were measured, as well as the robustness and performance of the generalists trained with them. New ways of measuring diversity were introduced to understand how the exploration versus exploitation optimization problem is best understood in collaborative contexts, as performance and robustness were both measured for generalists. By bettering collaborative capabilities of AI, especially for diverse human populations, AI can be applied to more everyday cooperative use cases. In addition, the new measures of diversity introduced in this study can be employed in the future to better understand artificially generated populations, and to better quantify the diversity of human strategies to accurately set goals for artificial diversity.

1. Introduction

1.1 The Human-Artificial Intelligence Cooperation Problem

Artificial Intelligence (AI) often struggles to cooperate with humans (Carroll et al., 2020). Formally, we define cooperation as a non-zero sum situation, where both the agent and the human are working towards a common goal (Bard et al., 2015). In a cooperative or non-zero-sum situation, what is good for one party is good for all parties (Bard et al., 2015).

This is an issue, as AI has become a much more prevalent technology in everyday life (Kennedy et al., 2023). The fitness, entertainment, transportation, security, customer service, (Kennedy et al, 2023) emergency response (Shuai et al., 2022), and healthcare (Shaw et al., 2019) industries have all in some way implemented AI, leading to more interactions between AI and humans.

1.2 Autonomous Vehicles as a Prime Example

Human-AI cooperative interactions can be awkward or difficult for AI to understand, as people can act unpredictably and suboptimally, and AI agents are often unable to react to such human actions. Although autonomous and semi-autonomous vehicles tend to get into fewer accidents than most drivers under normal conditions (Abdel-Aty and Ding, 2024), it is unlikely that the technology will be accepted by society until the accident rates are significantly better than that of human drivers.

To limit the number of these accidents, human-AI cooperation can be improved first in simpler environments, with lower risks, before implementation into high-risk environments, such as in autonomous vehicles. By understanding the concepts behind human-AI cooperation first, better algorithms can be implemented that are able to adapt to the wide range of human strategies that exist in high-risk environments, such as motorways (Aoki et al., 2021).

1.3 Human-AI Cooperation as a Challenge

Human-AI cooperation remains a challenge due to a few inherent differences between humans and traditional AI decision makers, formally known as agents (Nalepka et al., 2021). Firstly, traditional methods of training AI, such as Population Based Training (PBT), and Self Play (SP) tend to result in agents that perform well with each other by using pre-set conventions,

rather than adapting to their partners (Charakorn et al., 2020). These partners represent possible humans that the agent might encounter in an applied environment, so being able to adapt to this “simulated human” is imperative for an easy cooperative experience for the end user. While the strategies these methods tend to converge on are optimal in theory, they are only optimal because the agents can assume that their partners will also act optimally according to the same strategy, or policy (Zhao et al., 2022). However, humans rarely act in such a way, they are often unpredictable and suboptimal, necessitating that agents learn how to adapt to these suboptimal policies (Ribiero et al., 2022).

1.4 Solutions for Human-AI Cooperation

To address these issues, a variety of algorithms have been implemented to generate more robust agents, or agents capable of performing well with a wide variety of partners (Carroll et al., 2020) (Charakorn et al., 2022) (Charakorn et al., 2020) (Eysenbach et al., 2019) (Fontaine et al., 2021) (Rabinowitz et al., 2018) (Ribiero et al., 2022) (Wu et al., 2021) (Zhao et al., 2022).

1.4.1 Reinforcement Learning

This study, as well as many others, use Reinforcement Learning (RL) to create their generalists. RL is a field of AI which seeks to create an agent, or a decision-making algorithm, to accomplish a goal (Sutton et al., 2018). The performance of the agent in a run can be expressed as a number, where a better performance is a higher number, and a worse performance is a lower number (Sutton et al., 2018). This number is called the reward (Sutton et al., 2018). Thus, the goal of a reinforcement learning agent is to make decisions which maximize the reward (Sutton et al., 2018).

The way the agent makes decisions is by using a set of rules called a policy (Sutton et al., 2018). The agent is capable of taking observations, taking actions, and receiving rewards. After rewards are received, the policy is adjusted using a reinforcement learning algorithm to maximize the reward in the future, should the same observation, or a similar observation, be seen in the future (Sutton et al., 2018). The process is seen below in Figure 1.

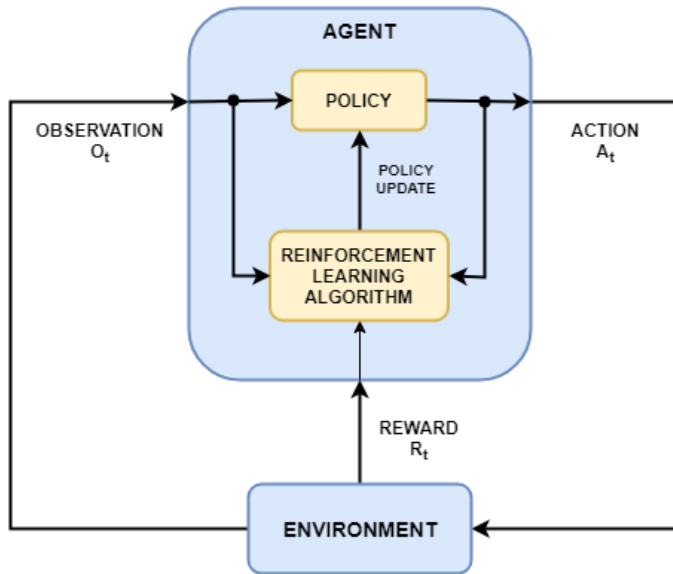


Figure 1: The Reinforcement Learning Cycle (What is Reinforcement, n.d.)

Figure 1 displays a simplified version of the reinforcement learning cycle. The agent is shown as both the policy and the reinforcement learning algorithm. The policy is capable of taking inputted observations, O_t about the state of the environment, and returning an action, A_t , which is the decision made based on the observation. The reinforcement learning algorithm is capable of receiving a reward, R_t , and updating the policy based on the reward to maximize R_t in the future (Deepmind, 2015).

The final part of the reinforcement learning cycle is the environment. The environment is the space the agent exists in. It takes A_t as an input, and based on the environment's state, returns O_t to the policy and R_t to the algorithm. A_t changes the state of the environment, or the condition of the environment, which determines O_t and R_t (Deepmind, 2015). Actions change the state of the environment (Deepmind, 2015).

One cycle takes one timestep, or t . In this timestep, the agent takes an action which may manipulate the state, then receives a reward and observation, which are used to change the policy and make the next decision. There are many timesteps in a single run, and the sum of the rewards accumulated throughout the run are the total rewards for the run. This total reward is used in other applications and methods (Deepmind, 2015).

1.4.2 Theory of Mind Reasoning

Rabinowitz et al., 2018 established an agent capable of Theory of Mind reasoning, or the ability to understand why another agent's actions were taken. A more applicable example of this reasoning is if a delivery truck is stopped in front of a human driver, the human driver can reason that the truck is stopped to deliver a package, and can act on this information by checking for a driver entering or exiting the truck before passing it. While this reasoning was a good proof of concept that cooperation was possible at a conceptual level, understanding the reasoning behind each action in a more complex environment, like the ones in the real world, was computationally impractical.

1.4.3 Population-Based Training

While Rabinowitz et al., 2018 approaches Human-AI cooperation as a meta-learning task, Zhao et al., 2022 leverages Population-Based Training (PBT) (Jaderberg et al., 2017) with an additional population entropy objective. PBT is an established method of creating agents, mimicking Darwinian evolution in a similar manner to Mariño et al., 2022, which further progressed in zero-sum (adversarial) environments. While traditional PBT results in non-robust agents by nature, by using an entropy bonus, Maximum Entropy Population-Based Training, or MEP, was able to produce a set of agents with different suboptimal strategies that are still somewhat effective, similar to possible human strategies (Zhao et al., 2022). Then, by using this set of MEP agents as training partners for an RL agent, a robust RL Generalist agent was able to be trained.

1.4.4 Learning Incompatible Strategies

Charakorn et al., 2022 uses a similar algorithm called Learning Incompatible Strategies (LIPO), which seeks to create diverse training populations by leveraging incompatibility. Charakorn et al., 2022 proves that incompatible strategies are dissimilar, thus, in a population, if

all the agents are incompatible with each other, all the agents are dissimilar, and so the population is diverse. By incentivizing incompatibility after each epoch by measuring for cross-play return across the set of agents in the population and adding a minimization objective to the training paradigm, the population becomes naturally more incompatible, while individual pairs within the population remain compatible. Thus, intrapair cooperation and similarity are high, and interpair cooperation and similarity are low for the LIPO training population. The training of these LIPO populations can be edited by changing the XPMIn, or Experience Minimization Objective, which controls the agent's balance between the dual objectives of diversity from other agents in the population and raw performance. By again using RL with these training partners from the LIPO method, a robust generalist can be created.

1.4.5 Environments

RL agents must be trained and tested in an environment that mimics real human-AI cooperation. To this end, a variety of environments have been created.

1.4.5.1 The Overcooked Environment

One popular cooperative environment is the Overcooked Environment, introduced in Carroll et al., which was modelled after the popular video game *Overcooked*. Figure 2 below shows a graphic of the Overcooked Environment.

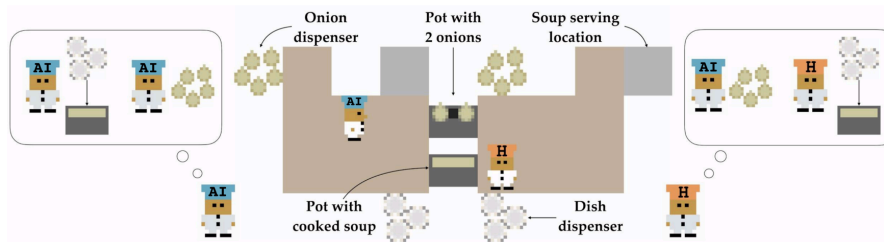


Figure 2, *The Overcooked Environment* (Carroll et al., 2020)

The Overcooked Environment features two players, usually a human or human-proxy with an agent, who cooperate. The goal of the agent and the human are to score points in a certain amount of time by delivering soup items to the soup serving location within the environment. To

do so, the agent and human must take onion items from the onion dispenser location and place them in pot locations. After a certain amount of time, the pot's state changes from holding onions to cooked soup, which can be interacted with a dish item to create a finished soup dish. This dish can be placed in the soup serving location to earn points (Carroll et al., 2020).

This environment is deceptively simple. Agents who undergo self-training often form stubborn behaviors and patterns depending on the layout they were trained on. This leads to inflexibility when performing with human partners, who often make unpredictable, suboptimal choices, leading to poor cooperation (Carroll et al., 2020).

The environment's layout heavily influences the most optimal strategy, and how complex coordination must be to succeed (Carroll et al., 2020). The layouts from Carroll et al. are shown below in Figure 3.

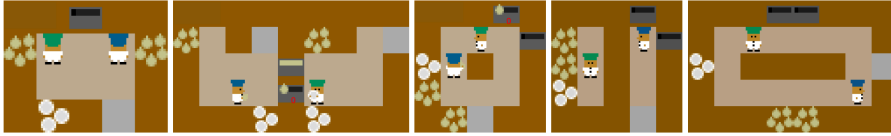


Figure 3, *Experimental Overcooked Layouts* (Carroll et al., 2020)

While Carroll et al. delves further into depth into the differences in each layout and the issues they present, it is evident that an environment such as the layout to the far left is less complex than the layout directly to the right of it (Carroll et al., 2020).

It should be noted that the Overcooked Environment, and the environments modelled after it, which are covered in the remainder of this section, are perfect information environments, where $O_t = S_t$ (Carroll et al., 2020). This makes it different imperfect information environments, where $O_t \neq S_t$ (Bard et al., 2015).

1.4.5.2 The Multi-Recipe Overcooked Environment

An extension to the Overcooked Environment is the Multi-Recipe Overcooked Environment as presented in Wu et al., which was extended in (Charakorn et al., 2022), which this paper uses. This Multi-Recipe Overcooked Environment, as the name states, features multiple possible recipes that can be made simultaneously from the same ingredients. It proves to be a much more

complex environment than the original Overcooked Environment for this reason. Figure 4 displays an overview of the Multi-Recipe Overcooked Environment as used in (Charakorn et al., 2022) including an example of an initial state and an overview of all possible recipes.

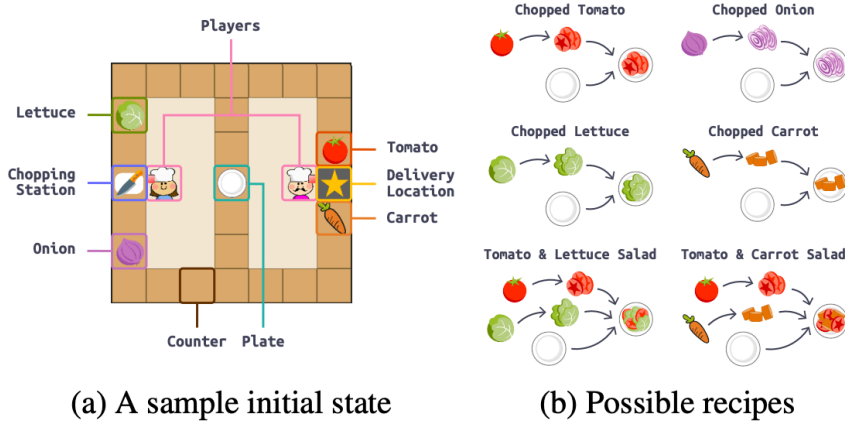


Figure 4, Overview of the Multi-Recipe Overcooked Environment (Charakorn et al., 2022)

It should be noted that Tomato and Carrot strategies are harder than those without these ingredients, since there is an additional step of passing the ingredients over the counter (Charakorn et al., 2022).

1.5 Limitations of Current Cooperative Agents

The balance of diversity and raw performance for training partners has not been found, leading to computational inefficiencies (Charakorn et al., 2022) (Eysenbach et al., 2018). Certain paradigms optimize too heavily for diversity, leading to agents that act differently from each other, but each agent performs too poorly to represent a semi-logical human strategy. Other paradigms optimize too heavily for performance, leading to partners that are too similar to one another, not allowing the generalist to understand a vaster array of possible strategies (Zhao et al., 2022), (Liu et al., 2021).

In addition, agents are often tested with agents from a held-out set for measuring performance (Charakorn et al., 2020). This helps to eliminate bias, but there is no common practice to determine if the testing set and the training set were similar. Lastly, diversity is often measured with difficult or proprietary methods, leading to minimal standardization of measure (Nalepka et al., 2021) (Charakorn et al., 2022) (Eysenbach et al., 2018).

1.6 Proposed Research

In short, the three problems identified herein within the field of cooperative agents are a lack of optimization between exploration and exploitation, a lack of confirmation of the dissimilarity of the testing and training populations, and the difficulty of measuring diversity of training or testing populations. This research sets out to create more optimized training partners for generalists to train with by understanding the effect of diversity objectives on performance and diversity. In addition, to support these findings, this research creates a method to confirm dissimilarity of the testing and training populations, and a simpler way to easily measure the diversity of a population

By modifying the XPMIn, I hypothesize that the diversity of a LIPO population can be modified, as XPMIn and diversity have a positive relationship. In addition, if a population is more diverse, I hypothesize that a generalist trained on that population should be more robust, and thus by modifying the XPMIn of a LIPO population during generation, a Generalist can be trained to be more robust, and cooperate better with a wider variety of partners, making it better suited for implementation in the real world, where human strategies are so diverse.

In addition, I hypothesize that if the XPMIn of a LIPO population is higher, then the overall performance of the agents will be lower, as the agents will prioritize diversity over performance.

Lastly, by measuring the diversity of a combined set of both the training and testing sets, I can determine any possible similarities and detect biases during training that may skew the performance of the generalist. By measuring diversity in a simple, easy way, I can institute this measure in future research, allowing for more standardization of diversity measures in the future.

2. Methods and Materials

Figure 5 below gives an overview for this section.

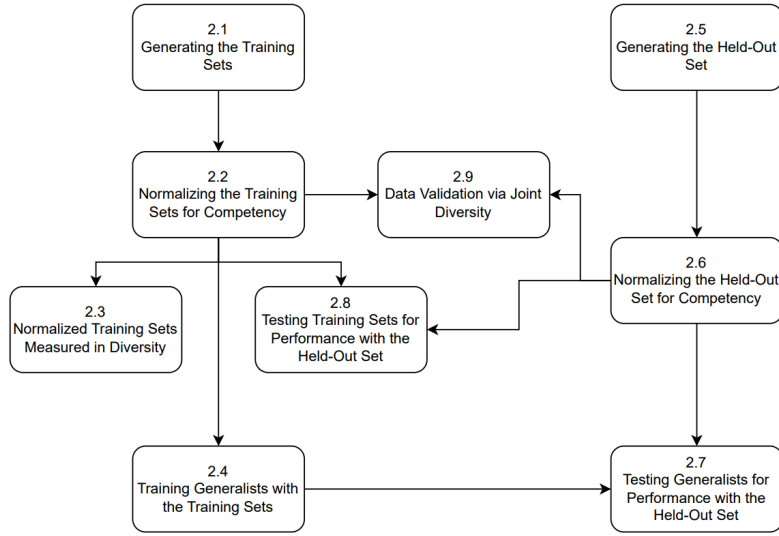


Figure 5, Overview of Methods

2.1 Generating the Training Sets

A total of 120 agents were generated using the LIPO method (Charakorn et al., 2022). These agents had one of five varying $XPM_{in} = \{0.1, 0.2, 0.3, 0.4, 0.6\}$, and for each XPM_{in} level, there were 3 batches of 8 agents generated.

2.2 Normalizing the Training Sets for Competency

Each agent was allowed to perform 3 runs of Self-Play (SP), in which the agent runs a testing round against itself. If the agent was not able to complete a single recipe across these 3 runs, it was removed from the set, normalizing the set to only include somewhat competent agents.

The nature of the LIPO method necessitates this normalization. Since in training, agents are motivated to not only perform well, but to act differently than each other, it is possible for an agent to finish training with a strategy that doesn't necessarily incentivize recipe completion, instead opting to perform seemingly random actions that are different than others in the set.

In practice, these SP rounds were incredibly easy to conduct, and any industry hoping to adopt LIPO should be able to easily implement this practice.

2.3 Normalized Training Sets Measured in Diversity

A novel diversity measure, SpaDiv was introduced to avoid the computational costs of most modern diversity measures.

Each agent A in a population P was allowed to play N rounds of SP, and each type of recipe that was completed was recorded. For each recipe, each agent could count at most 1 time, to prevent skewed data due to incredibly efficient SP agents that only complete one recipe. This recipe data was compiled, such that 6 counts C_r , one for each recipe r in the set of all recipes, were generated for each agent A in the population. Lastly, the Chi-Squared test of these 6 counts was calculated, and this number determined the diversity of population P.

Mathematically, SpaDiv for a population P of size n can be summarized as the following:

$$SpaDiv(P) = \frac{\sum_{A=A_0}^{A_n} X^2[C_{r,SP} | r \in \{R\}, SP = SP(A|A \in P)]}{n}$$

Equation 1, SpaDiv of a Population P

Each training set was measured using SpaDiv to determine the relationship between XPMin and diversity.

2.4 Training Generalists with Training Sets

Five generalists were produced via Reinforcement Learning with each batch of training partners after they were normalized for incompetency. All variables other than the number of agents produced and the XPMin were kept constant from Charakorn et al., 2022

2.5 Generating the Held-Out Set

A set of 12 LIPO agents was generated to act as testing partners for later steps. The XPMin was set to 0.3, and all other variables were kept constant.

2.6 Normalizing the Held-Out Sets for Competency

The same methods used in 2.2 were applied to the Held-Out Set to ensure that all agents within the set were competent. Since the Held-Out Set agents were to act as testing partners in

later steps, and we assume that all human strategies are competent (as defined by achieving at least one recipe in SP), it was reasonable to only test generalists with competent testing partners.

2.7 Testing Generalists for Performance with the Held-Out Set

Each generalist was allowed to play with each member of the Held-Out Set twice. The scores were recorded before data analysis was conducted to determine the relationship between XPMIn and final generalist performance.

2.8 Testing Training Sets for Performance with the Held-Out Set

Each competent member of the training sets was allowed to play with each member of the Held-Out Set twice, in the same manner as step 2.7. The scores were recorded before data analysis was conducted to determine the relationship between XPMIn and performance.

2.9 Data Validation via Joint Diversity

A Joint Set J_b as created of the Held-Out Set H and the generalist agents G_b , which were selected from the subset of all generalists G that were trained on the same batch b , which was selected from the set of all batches B . The SpaDiv of this joint set J was measured to determine the diversity of the set. By comparing the diversity $D(J_b)$ of J_b for each b , possible biases of H towards any one G_b could be detected, validating the data collected in steps 2.7 and 2.8.

The data validation method can be expressed mathematically with the following:

$$D(J_b) = \text{SpaDiv}[J_b | J_b = H + G_b, G_b \in (G | b, b \in B)]$$

Equation 2, Data Validation Method Using Joint Diversity

3. Data, Results, and Analysis

3.1 Diversity of Training Sets

Step 2.3 provided valuable information about the diversity of the training sets.

Table 1 below shows the diversity of training sets by XPMIn and Batch Number as measured by SpaDiv.

Table 1, Diversity of Batches by XPMIn

XPMIn	Batch 1	Batch 2	Batch 3	Mean	Standard Deviation
0.1	213.6854460	545.4125000	239.4500000	332.8493153	150.67246
0.2	159.6612903	423.4902507	291.5500000	291.5671803	107.70772
0.3	101.0	179.0632911	201.3666667	160.47665	43.03072
0.4	98.03448276	160.9113924	83.69565217	114.2138424	33.53502
0.6	80.0	112.3333334	144.8	112.3777778	26.454508

Table 1 allows for a qualitative judgement to be made that as XPMIn increases, the diversity of the batches decrease. This assertion seems counterintuitive, and so is confirmed through a quantitative analysis below.

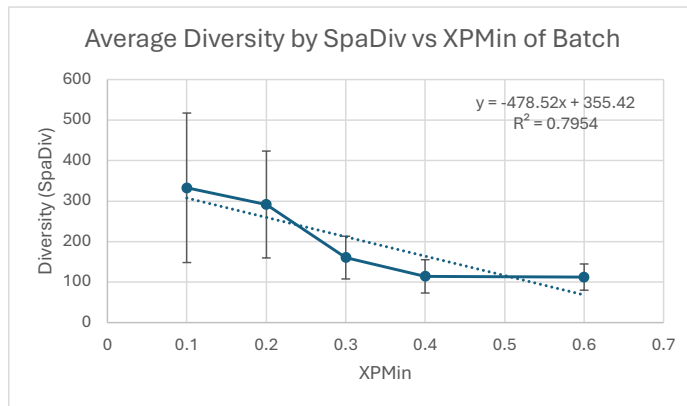


Figure 6, Average Diversity by SpaDiv vs XPMIn of Batch

Figure 6 demonstrates visually the inverse relationship between XPMIn and SpaDiv diversity. The linear regression shows a fairly strong negative correlation, with an R^2 value of 0.7954. This confirms the qualitative judgements made from Table 1, and would suggest that if XPMIn is raised, then the batches will be less diverse.

This statement, however, fails to consider Step 2.2, in which each batch was normalized for competency. Certain batches may have more agents than others, naturally leading to differing

diversities. While the batches may have been more diverse initially with higher XPMin, after normalization they have a lower quantity of agents, causing the population to be less diverse during testing.

3.1.1 Competency Rates

This hypothesis was tested by looking back at the data from Step 2.2, specifically into the quantity of agents left after the normalization. The quantity after normalization data is displayed below in Table 2.

Table 2, Competent Agents in Each Batch by XPMin

XPMin	Batch 1	Batch 2	Batch 3	Mean	Standard Deviation
0.1	8	8	4	6.6666667	1.8856181
0.2	5	6	6	5.66666667	0.47140452
0.3	4	4	6	4.66666667	0.88888889
0.4	3	4	2	3.00000000	0.81649658
0.6	2	2	3	2.33333333	0.47140452

Table 2 allows us to quantitatively notice the negative correlation between XPMin and the quantity of agents in each batch after normalization. However, this assertion still needs quantitative backing.

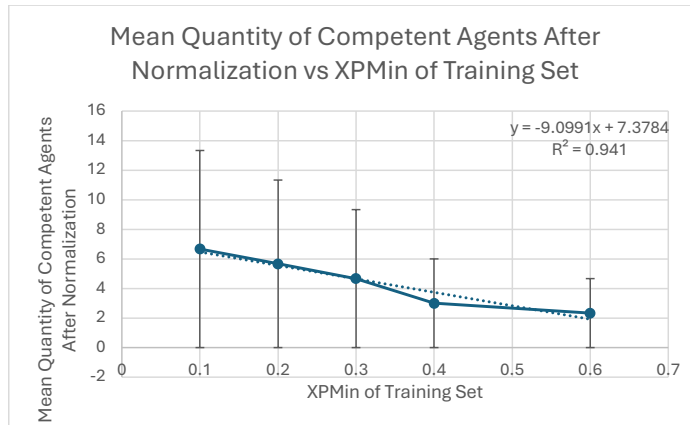


Figure 7, Mean Quantity of Competent Agents After Normalization vs XPMIn of Training Set

Figure 7 allows for a quantitative confirmation of the earlier statements via a linear regression. With an R^2 value of 0.874, a strong negative correlation is present between XPMIn and the number of agents in the batch after normalization. This explains the seemingly counterintuitive findings earlier that raising the XPMIn leads to lower diversity. While Charakorn et al., 2022 found that higher XPMIn leads to higher diversity, the batches tend to be less competent at these higher values, leading to smaller batches, which are naturally less diverse than larger batches.

3.2 Performance of Training Sets

The agents of the training sets, known as the Training Partners, were tested for performance after normalization, to see if there was a relationship between XPMIn and performance. While this finding isn't as applicable to generalists, it might prove useful in making more efficient use of computational resources in the future.

If training sets are normalized for diversity, it would make sense to have the greatest quantity of the most diverse possible agents. Thus, having agents that can perform competently, but not extremely well, would be optimal.

Commented [TS1]: How do I get the error bars to go negative?

Table 3, Performance of Training Partners by XPMIn

XPMIn	Mean	Standard Deviation
0.1	13.443504644064781	2.913941422434067
0.2	8.475857626719803	4.242639222891654
0.3	0.9823526623476223	5.647778800780726
0.4	-2.601470846189734	4.193776336198099
0.6	5.726715467742407	4.303081035385169

Table 3 above allows for a qualitative assertion that higher XPMIn batches perform poorly compared to low XPMIn batches. Figure X below confirms this statement graphically and quantitatively.

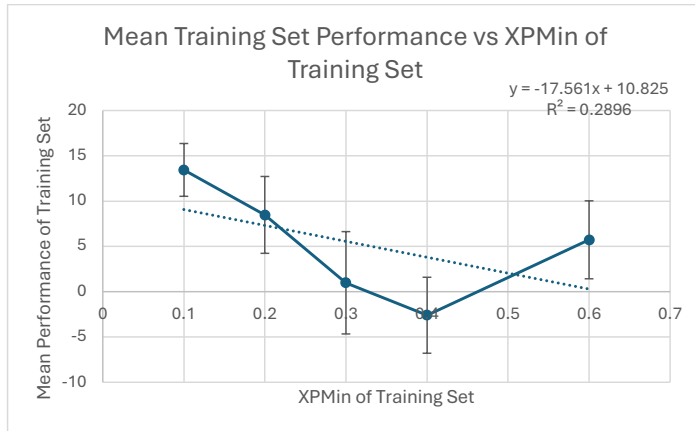


Figure 8, Mean Training Set Performance vs XPMIn of Training Set

Figure 8 clearly shows how XPMIn and performance have a negative correlation via a linear regression. While the R^2 value suggests a relatively weak relationship, this is likely the result of random noise with the 0.6 batches, as greater XPMIn tends to result in noisier batches (Charakorn et al., 2022).

3.3 Generalist Performance

Generalist performance is ultimately the most important metric of this paper. The primary goal of the field of Human-AI Cooperation is to create high-performing, robust generalists, and so the performance of the generalists with a variety of testing partners is the primary measure of progress in the field.

Table 4, Generalist Performance by Batch Used in Training

XPMin	Batch 1	Batch 2	Batch 3	Mean	Standard Deviation
0.1	13.78484977 1868437	13.53239976 4239788	13.54859978 1733007	13.62194977 2613744	0.1153774
0.2	12.78374974 1856008	13.33189976 3654917	10.43345810 0041996	12.183036	1.2572149
0.3	13.06194977 389276	12.90569974 378124	13.19359976 5267223	13.05374976 0980407	0.11767763

Table 4 above allows for a qualitative assessment of the performance of generalists trained with different training sets, showing that there is close to no relationship between performance and XPMin of the training set. Note that generalists were trained on sets of 0.4 or 0.6 XPMin due to time and computational constraints.

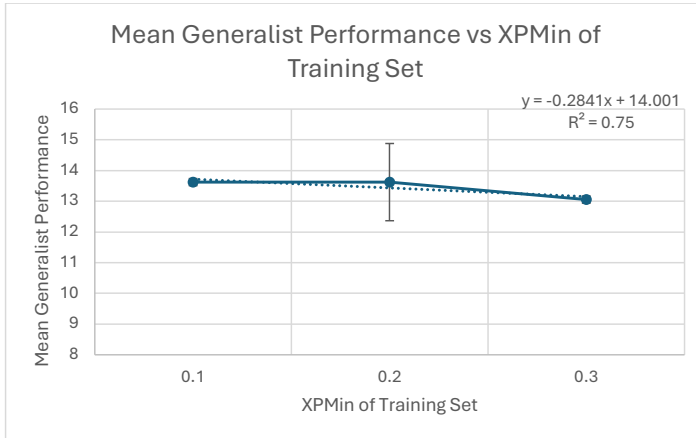


Figure 9, Mean Generalist Performance vs XPMIn of Training Set

Figure 9 above shows that there is a near zero relationship between XPMIn and performance via a linear regression. While extremely high XPMIn training sets were not used, it was already established in 3.1 and 3.2 that high XPMIn training sets are inefficient due to low competency and performance relative to lower XPMIn training sets, and so they can be discounted.

3.4 Data Validation

In order to validate the results of steps 3.2 and 3.3, step 2.9 was performed. The data collected from this step is in Table 5 below.

Table 5, Joint Diversity with Testing Set by Batch

XPMIn	Batch 1	Batch 2	Batch 3	Mean	Standard Deviation
0.1	166.6727273	263.3796791	165.6875	198.5799688	45.82208
0.2	102.2876712	210.2873563	218.8292683	177.1347653	53.03965
0.3	104.2857143	98.02068966	116.3239437	106.2101159	7.5951637
0.4	115.7692308	113.125	71.56250000	100.1522436	20.244803

0.6	62.94029851	118.0370370	129.8235294	103.6002883	29.150831
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Table 5 above qualitatively shows that the partners in the batches with a higher XPMIn happened to have similar strategies to the testing agents. This would suggest that the lower XPMIn partners had an unfair disadvantage, exacerbating the different in performance between the training sets themselves and the downstream generalists trained on those sets.

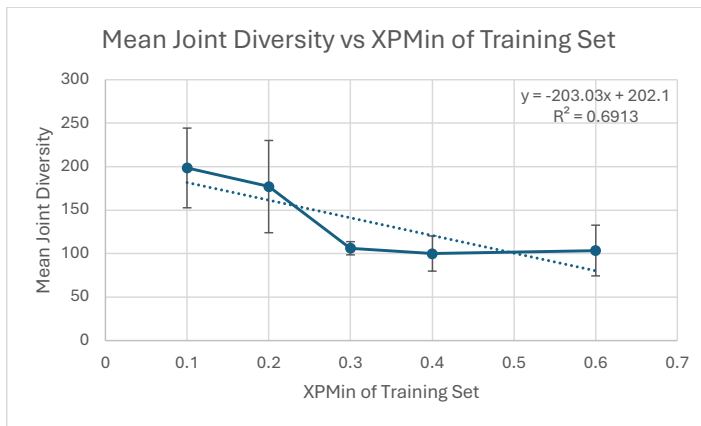


Figure 10, Mean Joint Diversity vs XPMIn of Training Set

Figure 10 above quantitatively confirms these results via a linear regression. Although the relationship is somewhat weak, there is still clearly a negative correlation with the XPMIn of the training set and the mean joint diversity of that set with the testing set.

4. Discussion and Future Research

4.1 Summary

This paper challenges the conventional XPMIn value of 0.3, presenting lower XPMIn values as a better alternative for more efficient production, higher performing partners, and higher performing generalists. It also presents SpaDiv, a more computationally efficient method of finding the diversity of a population of agents, relying only on self-play, a common function for most cooperative AI frameworks (Carroll et al., 2020), (Charakorn et al., 2022), (Bard et al.,

2020), (Eyesenbach et al., 2018), (Fontaine et al., 2021), (Liu et al., 2021), (Nalepka et al., 2021), (Wu et al., 2021), (Zhao et al., 2022).

4.2 Applications

While human-AI cooperation has a myriad of possible applications, it can be applied most directly at present to autonomous and semi-autonomous vehicles. As discussed in 1.2, the agents that drive these vehicles are still imperfect, and while they are better than humans in some cases, they still struggle under imperfect circumstances (Abdel-Aty and Ding, 2024).

More robust generalists to operate these vehicles can leverage not only more data, but also data-less such as Reinforcement Learning with diverse partners. This requires large populations of partners, and while computational power isn't as scarce in industry, it is still a limiting factor (Sastry et al., 2024). Having to discard large amounts of training partners due to incompetency would be wasteful, and so using multiple batches with lower XPMIn may be optimal to optimize diversity and reduce incompetency in industry.

4.3 Limitations

The chief limitation of this study was a lack of computational power. It necessitated SpaDiv, which while valuable, was not validated via another diversity measure. It also prevented the training of generalists with training sets of XPMIn larger than 0.3, and made it infeasible to generate training sets of more precise XPMIn levels.

In addition, this study focused on the Multi-Recipe Overcooked Environment, which has limited human data. Real applications, such as autonomous vehicles, shipping and transport, and personal assistance have large datasets about human behavior in those fields, which should be leveraged in order to allow for generalists to better understand human behavior in those environments.

Lastly, there isn't sufficient data about the actual diversity of human behavior across environments. While this seems obvious, as there is no feasible way to collect, organize, and analyze all human behavior, some benchmark would be important for the creation of diverse training sets. If training sets are able to mimic the level of diversity of human behavior, they

could help train generalists for real-world application, however this benchmark does not exist yet, so this is not possible.

4.4 Future Work

Immediately expanding upon this paper, more training sets should be created with varying XPMIn levels, preferably with more precise levels between the levels of this paper. This would not only confirm the results of this paper, which conflict with the conventions set in Charakorn et al., 2022, but seem counterintuitive at face value. In addition, SpaDiv should be confirmed with another diversity measure, such as EAV from McKee et al., 2022 to ensure that it is a viable method for measuring diversity.

Preparing this work for real-world application, it is necessary to understand how data-driven agents compare with data-less agents, such as the agents presented in this study. Likely, some agent combining the two methods, with a strategy first grounded with data-less partners before refinement through data is optimal.

Lastly, it is necessary to try producing training sets in larger quantities to see how different XPMins compare in computational efficiency. It is unknown whether the increased efficiency of producing competent partners at a lower XPMIn is worth the slight hypothesized dip in diversity at higher quantities of agents.